

Counterexamples to the Stallings Equalizer Conjecture via Colored Stallings Graphs

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Abstract

This report summarizes the main results presented in “*Colored Stallings Graphs and Counterexamples to the Stallings Equalizer Conjecture*” by Jialin Lei and Teng Zhang [1]. By introducing colored Stallings graphs, the authors construct an explicit family of counterexamples to the famous Stallings Equalizer Conjecture, proving that for every $n \geq 2$ there exist monomorphisms $g, h : F_n \rightarrow F_2$ such that $\text{rk Eq}(g, h) \geq 2n - 2$.

1 Introduction

An important problem in the study of free groups is to study the complexity of fixed point sets under homomorphisms between free groups. Let $F(\Sigma)$ be the free group generated by the set Σ . Given two homomorphisms $g, h : F(\Sigma) \rightarrow F(\Delta)$, their *equalizer* is defined by

$$\text{Eq}(g, h) := \{w \in F(\Sigma) : g(w) = h(w)\}.$$

In Stallings’ foundational work on automorphisms of free groups [2], he proposed his famous rank- n equalizer conjecture:

Conjecture 1.1 (Stallings, 1987). Let $n \geq 2$, let F_n be a free group of rank n , and let F be any free group. For any two monomorphisms $g, h : F_n \rightarrow F$,

$$\text{rk Eq}(g, h) \leq n.$$

Goldstein and Turner [3] proved that under the hypotheses of the conjecture, the equalizer is always finitely generated, and later they [4] showed that the equalizer still has finite rank under the weaker assumption that at least one of g and h is injective. The conjecture is also considered under such weaker assumption; see [5, Problem F31], [6, Problem 6] and [7, Conjecture 8.3]. Nevertheless, obtaining an upper bound on the rank remained elusive.

In the special case when $\Sigma = \Delta$ and $h = \text{id}$, the equalizer is precisely the fixed subgroup of the endomorphism g :

$$\text{Fix}(g) = \{w \in F(\Sigma) \mid g(w) = w\} = \text{Eq}(g, \text{id}).$$

Therefore, the fixed subgroup problem is closely related to the equalizer problem. The Scott Conjecture for automorphisms was proved by Bestvina and Handel [8] using train-track maps, and later Imrich and Turner [9] extended the result to all endomorphisms. Dicks and Ventura [6] proved stronger inertness results for families of injective endomorphisms, and Bergman [10] extended their results to families of endomorphisms.

For general equalizers, the situation is harder. Ventura [7] observed that the injectivity hypothesis is essential: without it, equalizers may have infinite rank. Ciobanu and Logan [11] established algorithmic and rank results for immersions, relating the problem to the Post Correspondence Problem. Logan [12] proved the case $n = 2$ in 2022, which remained the only completely settled case. For $n \geq 3$, the conjecture stood open for over 40 years.

The purpose of this paper is to give a family of explicit counterexamples to Conjecture 1.1 for $n \geq 3$. Our main result is the following.

Theorem 1.2. Let $n \geq 2$ and F_n be a free group of rank n . Then there exist monomorphisms $g, h : F_n \rightarrow F_2$ such that

$$\text{rk Eq}(g, h) \geq 2n - 2.$$

Consequently, Conjecture 1.1 is false for every $n \geq 3$.

2 Preliminaries: Stallings Graphs

Stallings graphs provide a powerful combinatorial tool for studying subgroups of free groups. We briefly recall the corresponding basic definitions and facts.

Definition 2.1 (X -labeled graph). Let X be a finite set, and let $F(X)$ be the free group with basis X . An X -labeled graph Γ consists of a vertex set $V\Gamma$, an oriented edge set $E\Gamma$, maps $\iota, \tau : E\Gamma \rightarrow V\Gamma$ recording the initial and terminal vertices, a fixed-point-free involution $e \mapsto \bar{e}$ on $E\Gamma$, and a labelling map $\ell : E\Gamma \rightarrow X^{\pm 1}$, such that

$$\iota(\bar{e}) = \tau(e), \quad \tau(\bar{e}) = \iota(e), \quad \ell(\bar{e}) = \ell(e)^{-1}.$$

Additional notes are as follows:

1. a *topological edge* is the unordered pair $\{e, \bar{e}\}$. We write $E^+\Gamma$ for a choice of one oriented edge from each topological edge;
2. an *oriented path* p in Γ is a finite sequence of oriented edges $p = e_1 e_2 \cdots e_m$ such that $\tau(e_j) = \iota(e_{j+1})$ for all j . Its label is $\ell(p) = \ell(e_1) \ell(e_2) \cdots \ell(e_m) \in F(X)$. The empty path has label 1. The inverse path is $\bar{p} = \bar{e}_m \cdots \bar{e}_1$;
3. a path is *reduced* if it has no immediate backtracking, that is, if $e_{j+1} \neq \bar{e}_j$ for all j ;
4. A *morphism* of based X -labeled graphs is a basepoint-preserving graph morphism which preserves initial vertices, terminal vertices, inverse edges, and labels;
5. if (Γ, v_0) is a connected based X -labeled graph, define the subgroup it represents as

$$L_X(\Gamma, v_0) = \{\ell(p) \in F(X) \mid p \text{ is a closed path in } \Gamma \text{ based at } v_0\}.$$

Definition 2.2 (Folded graph). An X -labeled graph Γ is called *folded* if for every vertex $v \in V\Gamma$ and every letter $x \in X^{\pm 1}$, there is at most one directed edge e such that $\iota(e) = v$ and $\ell(e) = x$.

Definition 2.3 (The Stallings graph of a subgroup). Let $H \leq F(X)$ be a subgroup.

- (1) The *Schreier covering graph* $\Sigma_X(H)$ has vertices the right cosets Hg for each $g \in F(X)$, base vertex H , and for each $x \in X$ a directed edge $Hg \xrightarrow{x} Hgx$. The reverse edge has label x^{-1} . Thus every reduced word $w \in F(X)$ determines a unique reduced path in $\Sigma_X(H)$ starting at the base vertex H , and this path ends at the vertex Hw . In particular, this path is closed at the base vertex if and only if $w \in H$.
- (2) The *based core subgraph* of $\Sigma_X(H)$ is the union of all reduced closed paths in $\Sigma_X(H)$ based at the vertex H . Equivalently, it is the subgraph consisting of all vertices and edges which occur along the path labeled by the reduced representative of some element of H . We denote this based subgraph by $\Gamma_X(H)$.
- (3) The above based graph $\Gamma_X(H)$ is called the *Stallings graph* of H over X . When the underlying set and subgroup are clear, we simply call it *Stallings graph*. By the former construction, $\Gamma_X(H)$ is connected and folded, and it represents exactly H :

$$L_X(\Gamma_X(H), *_H) = H,$$

where we denote the base vertex H by $*_H$. Moreover, $\Gamma_X(H)$ is finite if and only if H is finitely generated. In this case, the rank of the subgroup can be read directly from the graph via the Euler characteristic formula:

$$\text{rk } H = 1 - |V\Gamma_X(H)| + |E^+\Gamma_X(H)|,$$

where $E^+\Gamma_X(H)$ is a set of orientations, one for each topological edge.

Remark 2.4. The following facts will be used later:

- (1) As shown above, every subgroup of $F(X)$ has an associated based X -labeled graph. Conversely, let (Γ, v_0) be a connected folded based X -labeled graph. Suppose that every topological edge of Γ lies in some reduced closed path based at v_0 , then

$$(\Gamma, v_0) \cong \Gamma_X(L_X(\Gamma, v_0))$$

as based X -labeled graphs. Thus subgroups $H \leq F(X)$ are represented, up to based label-preserving isomorphism, by connected folded based X -labeled graphs in which every edge lies in a reduced closed path based at the base vertex.

- (2) If $A \leq B \leq F(X)$, then the map on right cosets $Ag \mapsto Bg$ induces a basepoint-preserving label-preserving morphism of Schreier graphs $\Sigma_X(A) \rightarrow \Sigma_X(B)$, which restricts to a natural morphism of Stallings graphs $\rho_{A,B} : \Gamma_X(A) \rightarrow \Gamma_X(B)$.
- (3) Recall that if Γ is a finite and connected graph, then

$$\text{rk } \pi_1(\Gamma) = 1 - |V\Gamma| + |E^+\Gamma|.$$

In particular, if $H \leq F(X)$ is finitely generated, then

$$\text{rk } H = 1 - |V\Gamma_X(H)| + |E^+\Gamma_X(H)|.$$

- (4) If Y is a connected based subgraph of a connected based graph Z , then the inclusion $Y \hookrightarrow Z$ induces a free factor inclusion $\pi_1(Y) \leq_{\text{ff}} \pi_1(Z)$. Indeed, one may choose a maximal tree in Y and extend it to a maximal tree in Z . The graph basis associated to the larger tree contains the graph basis associated to Y .

3 Colored Stallings Graphs

The main innovation of the paper is the introduction of **colored Stallings graphs**, which link Stallings graphs to a pair of homomorphisms (g, h) in a geometric way.

Definition 3.1 (Colored X -graph). Let G be a free group and let $g, h : F(X) \rightarrow G$ be two homomorphisms. A *colored X -graph* for (g, h) is a connected based X -labeled graph (Γ, v_0) together with a *color map* $C : V\Gamma \rightarrow G$ such that

1. $C(v_0) = 1$ (the base vertex has the identity color);
2. for every directed edge $e \in E\Gamma$,

$$g(\ell(e)) = C(\iota(e)) h(\ell(e)) C(\tau(e))^{-1}. \quad (3.1)$$

Definition 3.2 (Colored Stallings graph). If the underlying based graph (Γ, v_0) in Definition 3.1 is a Stallings graph, then (Γ, v_0, C) is called a *colored Stallings graph*.

The following lemma provides a method for computing the color map.

Lemma 3.3. Let (Γ, v_0, C) be a colored X -graph for (g, h) . Let p be a directed path from v_0 to a vertex v . Then

$$C(v) = g(\ell(p))^{-1} h(\ell(p)).$$

Consequently, the label of every closed path in Γ based at v_0 belongs to $\text{Eq}(g, h)$.

Proof. The proof proceeds by induction on the length of the path. For the empty path the statement is clear since $C(v_0) = 1 = g(1)^{-1} h(1)$. Suppose the statement holds for a path p ending at v , and let e be an edge with $\iota(e) = v$, $\tau(e) = w$, and $\ell(e) = x$. Then by the coloring condition (3.1),

$$C(w) = g(x)^{-1} C(v) h(x) = g(x)^{-1} g(\ell(p))^{-1} h(\ell(p)) h(x) = g(\ell(p)x)^{-1} h(\ell(p)x).$$

□

The following lemma shows that the injectivity of the color map allows us to construct a subgroup of the equalizer via a colored Stallings graph, which yields a lower bound on the rank of the equalizer. Therefore, the problem of bounding the rank of the equalizer is reduced to a combinatorial construction on graphs.

Lemma 3.4. Let (Γ, v_0, C) be a finite colored Stallings graph for (g, h) . Assume that the color map $C : V\Gamma \rightarrow G$ is injective. Let $A = L_X(\Gamma, v_0) \leq F(X)$. Then

$$A \leq \text{Eq}(g, h),$$

and the natural morphism

$$\rho : \Gamma_X(A) \longrightarrow \Gamma_X(\text{Eq}(g, h))$$

is injective on vertices and on edges. Consequently,

$$A \leq_{\text{ff}} \text{Eq}(g, h).$$

In particular,

$$\text{rk } \text{Eq}(g, h) \geq \text{rk } A.$$

Proof. By Lemma 3.3, $A \leq \text{Eq}(g, h)$. Since (Γ, v_0) is a Stallings graph, Remark 2.4 gives that $(\Gamma, v_0) \cong \Gamma_X(A)$. Therefore, the inclusion $A \leq \text{Eq}(g, h)$ induces a natural map $\rho : \Gamma_X(A) \rightarrow \Gamma_X(\text{Eq}(g, h))$. Let $u, v \in V\Gamma$, and suppose that $\rho(u) = \rho(v)$. Choose oriented paths p_u, p_v in Γ from v_0 to u and v , respectively. Thus, the path $\rho(p_u)\overline{\rho(p_v)}$ is a closed path based at the base vertex of $\Gamma_X(\text{Eq}(g, h))$. Consequently,

$$\ell(p_u)\ell(p_v)^{-1} \in \text{Eq}(g, h).$$

Therefore

$$g(\ell(p_u)\ell(p_v)^{-1}) = h(\ell(p_u)\ell(p_v)^{-1}).$$

By Lemma 3.3, we can easily obtain that $C(u) = C(v)$. Since C is injective, it follows that $u = v$. Since Γ is folded, by vertex injectivity, ρ is also injective on edges. $\square$

4 Construction of Counterexamples

The counterexample is constructed in two steps: first defining the homomorphisms g and h , and then building a colored Stallings graph that induces a large free factor of the equalizer.

4.1 Construction of the Homomorphisms

Let $X_n = \{t, x_1, \dots, x_{n-1}\}$, and let $F_n = \langle t, x_1, \dots, x_{n-1} \rangle$ be a free group of rank n . For $0 \leq i \leq n-1$, define

$$c_i = a^{-i}b^i.$$

We shall define

$$g(t) = a, \quad h(t) = b, \quad g(x_i) = h(x_i) = c_i^2.$$

For $1 \leq i \leq n-1$, set

$$d_i = b^i a^{-i} b^i.$$

We now prove that g and h are monomorphisms.

Lemma 4.1. $\{a, d_1, \dots, d_{n-1}\}$ freely generates a free subgroup of rank n .

Proof. We prove by using the Stallings graph of $\langle a, d_1, \dots, d_{n-1} \rangle \leq F(a, b)$. Define the graph Δ_n as follows. The graph Δ_n has a base vertex o with one a -loop. From o , there is a positive b -path

$$o = P_0 \xrightarrow{b} P_1 \xrightarrow{b} \dots \xrightarrow{b} P_{n-1}$$

and a negative b -path

$$o = Q_0 \xrightarrow{b^{-1}} Q_1 \xrightarrow{b^{-1}} \dots \xrightarrow{b^{-1}} Q_{n-1}.$$

For each $i = 1, \dots, n-1$, the vertices P_i and Q_i are connected by a path

$$P_i = R_{i,0} \xrightarrow{a^{-1}} R_{i,1} \xrightarrow{a^{-1}} \dots \xrightarrow{a^{-1}} R_{i,i} = Q_i$$

of length i . Let $X = \{a, b\}$. One verifies that Δ_n is the Stallings graph of $\langle a, d_1, \dots, d_{n-1} \rangle$ and $L_X(\Delta_n, o) = \langle a, d_1, \dots, d_{n-1} \rangle$ has rank n . $\square$

Proposition 4.2. The homomorphisms g and h defined above are monomorphisms.

Proof. We first prove that g is injective. For each i , put $y_i = t^i x_i$. Hence, $\{t, y_1, \dots, y_{n-1}\}$ is also a free basis of F_n . Note that $g(y_i) = d_i$. Thus, the image of $\{t, y_1, \dots, y_{n-1}\}$ under g is $\{a, d_1, \dots, d_{n-1}\}$. By Lemma 4.1, $\{a, d_1, \dots, d_{n-1}\}$ freely generates a free subgroup of rank n . Thus, g is injective.

We now prove that h is injective. Since g is injective, the set $\{a, c_1^2, \dots, c_{n-1}^2\}$ is a free basis of a group of rank n . Hence $\{a, c_1^{-2}, \dots, c_{n-1}^{-2}\}$ is also a free basis. Let

$$\sigma : F(a, b) \rightarrow F(a, b), \quad \sigma(a) = b, \quad \sigma(b) = a.$$

Therefore $\sigma(\{b, c_1^2, \dots, c_{n-1}^2\}) = \{a, c_1^{-2}, \dots, c_{n-1}^{-2}\}$. Thus, $\{b, c_1^2, \dots, c_{n-1}^2\}$ freely generates a free subgroup of rank n . Hence, h is injective. $\square$

4.2 Construction of the Colored Stallings Graph

We now construct the colored Stallings graph Γ_n that realizes the lower bound in Theorem 1.2 with n :

- Vertices: $v_0, v_1, \dots, v_{n-1}$ (total n vertices), with base vertex v_0 ;
- For each $1 \leq i \leq n-1$, a t -edge $v_{i-1} \xrightarrow{t} v_i$;
- For each $1 \leq i \leq n-1$, an x_i -loop at v_0 and an x_i -loop at v_i .

The color map is defined by $C(v_i) = c_i = a^{-i} b^i$.

It is straightforward to check that the triple (Γ_n, v_0, C) is a colored Stallings graph for the pair (g, h) . Further, the color map C is injective. Let $A_n = L_{X_n}(\Gamma_n, v_0) \leq F_n$, then $\text{rk } A_n = 1 - n + (3n - 3) = 2n - 2$.

Proof of Theorem 1.2. By Proposition 4.2, both g and h are monomorphisms. By Lemma 3.4, the subgroup $A_n = L_{X_n}(\Gamma_n, v_0)$ satisfies $A_n \leq_{\text{ff}} \text{Eq}(g, h)$ and $\text{rk } A_n = 2n - 2$. Hence $\text{rk } \text{Eq}(g, h) \geq 2n - 2$. If $n \geq 3$, then $2n - 2 > n$, so Conjecture 1.1 is false for every $n \geq 3$. $\square$

5 Further Discussion

5.1 Stabilized Non-Injective Applications

The construction above also yields consequences when one allows non injective maps. The reason is that one may add dummy generators without changing the large subgroup already lying in the equalizer.

Proposition 5.1. For every integer $r \geq 5$, there exist homomorphisms $\alpha, \beta : F_r \rightarrow F$ to a free group F such that α is injective, β is not injective, and $\text{rk } \text{Eq}(\alpha, \beta) > r$.

Sketch of Proof. Put $m = r - 1$. Let $X_m = \{t, x_1, \dots, x_{m-1}\}$ and $X_r = X_m \cup \{z\}$. Apply Theorem 1.2 to obtain monomorphisms $g, h : F_m \rightarrow F_2$ and the colored Stallings graph (Γ_m, v_0, C) whose represented subgroup $A_m = L_{X_m}(\Gamma_m, v_0)$ satisfies $A_m \leq_{\text{ff}} \text{Eq}(g, h)$ and $\text{rk } A_m = 2m - 2$.

Write $F_r = F_m * \langle z \rangle$ and let s be a new free generator with target $F = F_2 * \langle s \rangle$. Define $\alpha|_{F_m} = g$, $\alpha(z) = s$, and $\beta|_{F_m} = h$, $\beta(z) = 1$. Then α is injective, while β is not injective since $z \in \ker \beta$.

The X_r -Stallings graph of A_m is still (Γ_m, v_0) . The same coloring C makes (Γ_m, v_0, C) a colored X_r -Stallings graph for (α, β) , and the color map remains injective. Hence $A_m \leq_{\text{ff}} \text{Eq}(\alpha, \beta)$, and $\text{rk Eq}(\alpha, \beta) \geq \text{rk } A_m = 2m - 2 = 2r - 4$. For $r \geq 5$, one has $2r - 4 > r$. $\square$

Proposition 5.2. For every integer $r \geq 7$, there exist non-injective homomorphisms $\alpha, \beta : F_r \rightarrow F$ to a free group F such that $\text{rk Eq}(\alpha, \beta) > r$.

Sketch of Proof. Put $m = r - 2$. Let $X_m = \{t, x_1, \dots, x_{m-1}\}$ and $X_r = X_m \cup \{z_1, z_2\}$. Apply Theorem 1.2 to obtain monomorphisms $g, h : F_m \rightarrow F_2$ and the colored Stallings graph (Γ_m, v_0, C) with represented subgroup $A_m = L_{X_m}(\Gamma_m, v_0)$ satisfies $A_m \leq_{\text{ff}} \text{Eq}(g, h)$ and $\text{rk } A_m = 2m - 2$.

Write $F_r = F_m * \langle z_1 \rangle * \langle z_2 \rangle$ and let s be a new free generator with target $F = F_2 * \langle s \rangle$. Define $\alpha|_{F_m} = g$, $\beta|_{F_m} = h$, and set $\alpha(z_1) = 1$, $\beta(z_1) = s$, $\alpha(z_2) = s$, $\beta(z_2) = 1$. Then neither α nor β is injective.

As in Proposition 5.1, the X_r -Stallings graph of A_m is still (Γ_m, v_0) . The same coloring C makes (Γ_m, v_0, C) a colored X_r -Stallings graph for (α, β) , and the color map is still injective. Hence $A_m \leq_{\text{ff}} \text{Eq}(\alpha, \beta)$, and $\text{rk Eq}(\alpha, \beta) \geq \text{rk } A_m = 2m - 2 = 2r - 6$. For $r \geq 7$, one has $2r - 6 > r$. $\square$

These propositions show that even under the weaker assumption that only one or none of the maps are injective, the rank of the equalizer can still exceed n . In fact, once both maps are allowed to be non-injective, the equalizer may even fail to be finitely generated.

5.2 A Possible Sharp Upper Bound

The construction of counterexamples naturally raises the question: what is the correct upper bound? Based on the explicit examples, the authors propose the following revised conjecture:

Conjecture 5.3. Let $n \geq 3$, and let $g, h : F_n \rightarrow F$ be monomorphisms into a free group. Then

$$\text{rk Eq}(g, h) \leq 2n - 2.$$

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